

Flat Exit-Resampling (d'Eon & Křivánek)

TL;DR:

1. Exit Probability Distributions

The exit probability density function (ePDF) and cumulative distribution function (eCDF) for optical depth x and exit direction cosine u are defined as:

$$\text{ePDF}(x, u) = \frac{e^{x + \frac{x}{u}}}{1 - e^x x E_1(x)}$$

$$\text{eCDF}(x, u) = \frac{-1 - u e^{x + \frac{x}{u}} + e^x x E_1(x) - e^x x E_1\left(-\frac{x}{u}\right)}{-1 + e^x x E_1(x)}$$

Variables & Functions

- $u \in [-1, 0]$: Cosine of the exit direction angle relative to the surface normal.
- $x > 0$: Optical depth within the medium.
- $E_1(z) = \Gamma(0, z) = \int_1^\infty \frac{e^{-zt}}{t} dt$: Exponential integral function.

2. Weight Correction Derivation

When replacing standard directional sampling with the exit-resampled distribution $\text{ePDF}(x, u)$, an importance sampling weight correction factor $W(x, u)$ must be applied to maintain an unbiased Monte Carlo estimator.

1. **Original Directional Distribution:** The base directional distribution relative to solid angle measure is $p_{\text{orig}}(u)$.

2. **Exit-Resampled Distribution:** Directions $u \in [-1, 0]$ are generated according to:

$$p_{\text{exit}}(u) = \text{ePDF}(x, u)$$

3. Weight Correction Factor: The Monte Carlo weight adjustment factor is the ratio of the original target distribution to the sampling PDF:

$$W(x, u) = \frac{p_{\text{orig}}(u)}{\text{ePDF}(x, u)} = p_{\text{orig}}(u) \cdot \left[\frac{1 - e^x x E_1(x)}{e^{x + \frac{x}{u}}} \right]$$

Simplifying the exponential terms gives:

$$W(x, u) = p_{\text{orig}}(u) \cdot (1 - e^x x E_1(x)) e^{-x(1 + \frac{1}{u})}$$

Note: $W(x, u)$ alone corrects only for the change of *directional* sampling measure — it implicitly assumes escape from this bounce is *certain*. See Section 2.5 for the correction this omits.

2.5. Escape-Probability Correction

Section 2's $W(x, u)$ implicitly assumes the walk is *certain* to escape from this bounce. In practice, resampling is invoked speculatively: the walk only escapes from any given bounce with some probability, governed by the mixture of classical and Dwivedi-guided sampling actually in use. Applying $W(x, u)$ alone, without correcting for this, silently reintroduces bias.

The complete, unbiased per-sample weight is:

$$W_{\text{final}}(x, u_r) = \frac{\text{trans}(u_r)}{\text{ePDF}(x, u_r) \cdot P_{\text{esc, mix}}(x, \nu_0, \gamma)}$$

where $\text{trans}(u_r)$ is the real physical transmittance along the resampled exit direction, and γ is the guided fraction (the mixture weight between classical and Dwivedi-guided sampling actually used by the walk).

Escape probability under the classical scheme:

$$P_{\text{esc, classical}}(x) = \frac{1}{2} E_2(x), \quad E_2(x) = e^{-x} - x E_1(x)$$

Escape probability under the mixture actually used:

$$P_{\text{esc, mix}}(x, \nu_0, \gamma) = (1 - \gamma) P_{\text{esc, classical}}(x) + \gamma P_{\text{esc, Dwivedi}}(x, \nu_0)$$

using $P_{\text{esc,Dwivedi}}$ from Section 3.

Why both terms are necessary: $\text{ePDF}(x, u)$ is the correct direction density *conditional on escape*; $P_{\text{esc,mix}}$ is the probability that escape actually occurs at all under the original (un-resampled) sampling scheme, at this depth. By Bayes' rule, $P(\text{escape}, u) = P(\text{escape}) \cdot P(u \mid \text{escape})$ – so the estimator's denominator needs both factors, not just the conditional density. Omitting $P_{\text{esc,mix}}$ is equivalent to assuming $P(\text{escape}) = 1$, which is false whenever the medium absorbs.

3. Implementation Details

- **Analytical Closed Form for $P_{\text{esc,Dwivedi}}$ (replacing numerical quadrature):** Direction sampling itself (inverting $\text{eCDF}(x, u)$ for u) has no known closed form and is still performed via a safeguarded, bisection-fenced Newton-Raphson solver. What *does* admit a closed form is the separate scalar quantity $P_{\text{esc,Dwivedi}}(x, \nu_0)$ used in the escape-probability weight correction (Section 2.5) – originally computed via numerical quadrature, now evaluated directly via the substitution $w = -\frac{1}{u}$ and partial fraction decomposition on $\frac{1}{w(\nu_0 w - 1)} = -\frac{1}{w} + \frac{\nu_0}{\nu_0 w - 1}$:

$$P_{\text{esc,Dwivedi}}(x, \nu_0) = \frac{1}{\ln\left(\frac{\nu_0+1}{\nu_0-1}\right)} \exp\left(\frac{x}{\nu_0}\right) \left[-E_1(x) + \exp\left(-\frac{x}{\nu_0}\right) E_1\left(x \cdot \frac{\nu_0-1}{\nu_0}\right) \right]$$

Verified against direct numerical quadrature of the original integral definition of $P_{\text{esc,Dwivedi}}$ to within $\sim 10^{-31}$ (arbitrary precision) across $x \in \{0.2, 1, 2, 5\}$, $\nu_0 \in \{1.5, 2, 5\}$.

- **Exponential Integral Approximation:** For efficient evaluation in C++ shaders, piecewise rational approximations (Abramowitz & Stegun §5.1.53 / §5.1.56) process the $E_1(z)$ components to maintain a relative accuracy within 10^{-7} to 10^{-6} across the full numerical domain without performance bottlenecks.

4. Zero-Variance Conditional on Escape

The paper (§6.6.2) measures exit-resampling's variance reduction **empirically** – it never claims or derives an exact result for the Dwivedi/asymptotic-guiding scheme, since the underlying importance function ($W(x) = e^{-x/\nu_0}$) is itself only an approximation (the discrete asymptotic term, with the transient portion discarded). This section is a stronger, **proven** result about one piece of that approximate scheme.

Claim: the ratio of transmittance to sampling density along the resampled exit direction is *constant*, independent of u :

$$\frac{\text{trans}(u)}{\text{ePDF}(x, u)} = E_2(x), \quad \forall u \in [-1, 0]$$

Derivation. The classical free-flight transmittance kernel is $\text{trans}(u) = e^{x/u}$. Substituting the ePDF definition from Section 1:

$$\frac{\text{trans}(u)}{\text{ePDF}(x, u)} = \frac{e^{x/u}}{e^{x+x/u}} = \frac{1 - e^x x E_1(x)}{e^x} = e^{-x} - x E_1(x) = E_2(x)$$

The u -dependence cancels exactly — the ratio depends only on optical depth x , not on which exit direction was drawn. (*Verified numerically to 15+ significant digits.*)

Consequence. Since every resampled sample scores the identical value $E_2(x)$ whenever escape occurs, the resampled estimator is **exactly zero-variance conditional on escape** — the direction step contributes no randomness to the outcome at all. All remaining variance comes from the single Bernoulli "did it escape" gate governed by $P_{\text{esc},\text{mix}}(x, \nu_0, \gamma)$ from Section 2.5. This yields a clean closed form:

$$\text{Var}[\text{resampled}] = E_2(x)^2 \cdot \frac{1 - P_{\text{esc},\text{mix}}(x, \nu_0, \gamma)}{P_{\text{esc},\text{mix}}(x, \nu_0, \gamma)}$$

which is exactly the variance of a constant $c = E_2(x)$ scored with probability $p = P_{\text{esc},\text{mix}}$ and reweighted by $1/p$ for an unbiased estimator of $p \cdot c$: $\text{Var} = c^2 (1 - p)/p$.